Large-Scale Motor Insurance Risk Intelligence: Expected Claim Cost Prediction, Risk Classification, and Interactive Dashboards

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Abstract - This report presents an end-to-end motor insurance analytics project built from two connected dataset phases. Phase 1 uses the midterm insurance CSV files: data_synthetic_.csv with 54,503 customer records and 30 fields, and insurance_dataset.csv with 13,000 records and 7 fields. These files support the original customer segmentation, premium/claim exploration, and dashboard baseline. Phase 2 extends the project with the real freMTPL2 French motor third-party liability dataset for large-scale actuarial modeling. For the freMTPL2 pipeline, freMTPL2sev claim amounts are aggregated at policy level and left joined to freMTPL2freq, preserving zero-claim policies needed for actuarial modeling. The final modeling file contains 678,013 policies, 23 columns, 358,499.45 exposure units, 36,102 recorded claims, and 59.91 million in total claim amount. EDA shows a realistic insurance pattern: 94.98% of policies have no recorded claim, while a small number of policies produce very large losses. Expected claim cost is modeled using a mean baseline, Poisson frequency regression, Gamma severity regression, a frequency-severity product model, Tweedie pure-premium regression, and histogram gradient boosting. Histogram gradient boosting gives the lowest RMSE of 30,373.58, while Tweedie regression gives the lowest MAE of 584.02. A Random Forest classifier is also used in the dashboard to classify interpretable risk profiles derived from BonusMalus bands. The dashboards turn the technical work into a simple decision-support story for pricing, monitoring, and underwriting.

Index Terms - Insurance analytics, midterm insurance dataset, freMTPL2, pure premium, claim frequency, claim severity, BonusMalus, Random Forest, Tweedie regression, Power BI, Streamlit.

I. INTRODUCTION

Insurance companies need to estimate claim cost before they make pricing and portfolio decisions. The final price paid by a customer is not only expected loss; it also includes expenses, taxes, profit, market strategy, and regulation. For that reason, this report uses pure premium, or expected claim cost per exposure unit, as the technically correct target when actual charged premium is not available.

The project builds directly on the midterm work, so the midterm CSV files are part of the complete project history and evidence. In Phase 1, data_synthetic_.csv and insurance_dataset.csv were used to explore insurance customers, premium-related fields, claim amount behavior, segmentation groups, and dashboard storytelling. In Phase 2, the project adds freMTPL2 because it provides a larger and more actuarially realistic policy-claim structure for expected claim cost modeling. The report is written as an insurance analytics paper rather than a generic data science report. The story moves from raw data, to processing, to EDA, to ML results, to dashboards.

The project addresses three gaps from the earlier work without ignoring the midterm CSVs. First, it keeps the midterm datasets as the Phase 1 segmentation and dashboard foundation. Second, it adds freMTPL2 as the Phase 2 real-world actuarial dataset, giving 678,013 policy records for stronger ML. Third, it connects the model outputs to dashboards so the results are understandable for both technical and business audiences.

II. DATASET AND DATA PROCESSING

Dataset Source and Structure

The complete project uses three CSV inputs across two phases. The first midterm file, data_synthetic_.csv, contains 54,503 customer records and 30 columns, including customer demographics, occupation, income, policy dates, claim history, insurance products owned, coverage amount, premium amount, deductible, policy type, customer preferences, risk profile, previous claims history, credit score, driving record, life events, and segmentation group. This file is useful for customer segmentation and dashboard design because it contains premium and customer-behavior fields.

The second midterm file, insurance_dataset.csv, contains 13,000 records and 7 columns: Age, Gender, Income, Marital_Status, Education, Occupation, and Claim_Amount. This file is smaller and simpler, so it is used as supporting insurance reference data rather than the main final modeling table.

The final actuarial modeling dataset is freMTPL2, a public French motor third-party liability insurance dataset distributed through CASdatasets and public mirrors. It contains two linked tables. The frequency table has one row per policy and includes IDpol, ClaimNb, Exposure, VehPower, VehAge, DrivAge, BonusMalus, VehBrand, VehGas, Area, Density, and Region. The severity table contains claim amounts connected by IDpol.

Dataset	Columns	Rows	Role in project
data_synthetic_.csv	54,503	30	Midterm customer segmentation, premium fields, risk profile, and dashboard baseline.
insurance_dataset.csv	13,000	7	Supporting insurance claim-amount reference dataset.
freMTPL2freq + freMTPL2sev	678,013 policies	Final 23 columns	Final real-world expected claim cost / pure premium modeling dataset.

TABLE I. Dataset inventory used across the midterm and final project phases.

The data-quality checks also explain how each dataset was used. The midterm synthetic customer file has 2,521 missing values and 1,000 duplicate rows, so it is useful for Phase 1 segmentation and dashboard storytelling but not ideal as the only final modeling file. The smaller insurance_dataset.csv has no missing values or duplicate rows, but it has only seven columns and does not contain the same policy-exposure structure needed for actuarial pure-premium modeling. The freMTPL2 processed file is therefore the final cleaned and merged modeling dataset.

The raw frequency file contains 678,013 policies. The raw severity file contains 26,639 claim amount rows. After processing, the final modeling file still contains 678,013 policies and 23 columns. Keeping the full policy population is important because most policies have no claims, and removing them would create a biased claims-only dataset.

Processing Steps

The severity table was aggregated by IDpol to create TotalClaimAmount and SeverityClaimRows. This policy-level severity table was left joined to the frequency table. Missing joined claim amounts were set to zero, exposure was required to be positive, and target fields were engineered. The main target is PurePremium = TotalClaimAmount / Exposure.

Additional engineered fields include Frequency, HasClaim, AvgClaimAmount, SeverityObserved, ClaimCountMismatch, DriverAgeBand, VehicleAgeBand, and BonusMalusBand. The processing audit found 34,060 policies with ClaimNb greater than zero, 24,944 policies with positive joined claim amount, and 9,117 claim-count mismatch rows. These mismatch rows were retained and documented because real operational data often contains imperfect table alignment.

The final portfolio has 358,499.45 exposure units, 36,102 recorded claims, and 59,909,216.50 in total claim amount. The exposure-weighted claim frequency is 0.100703, and the exposure-weighted pure premium is 167.1110.

III. EXPLORATORY DATA ANALYSIS

EDA shows that the dataset is large, real, and naturally imbalanced. Out of 678,013 policies, 643,953 have zero recorded claims. This means 94.98% of the policies are zero-claim observations. At the same time, claim severity is heavy-tailed, so a small number of policies create very large claim costs. This is exactly the type of structure expected in insurance data.

The portfolio dashboard in Figure 1 summarizes the final freMTPL2 modeling population: 678,013 policies, 36,102 total recorded claims, and weighted pure premium of 167.11. There are two claim-rate concepts in the project. The policy-level claim incidence is 5.02%, calculated as policies with ClaimNb greater than zero divided by all policies. The exposure-weighted claim frequency is 0.100703 claims per exposure unit, calculated as total claim count divided by total exposure.

Driver age is a clear signal. Figure 4 shows claim frequency by driver age band, while the dashboard screenshots summarize policy-level claim incidence. Drivers aged 18-25 have the highest policy claim rate at 6.74%, compared with 4.23% for ages 26-35. Drivers aged 66+ also show an elevated policy claim rate of 5.91%. On an exposure-weighted basis, the 18-25 group has 0.1751 claims per exposure unit and a pure premium of 758.40, making it the highest-cost age band.

BonusMalus is the strongest EDA risk signal. The BonusMalus risk dashboard, and the dashboard view shows policy claim incidence rising from 4.49% for BonusMalus <=60 to 18.92% for BonusMalus 176+. The middle bands also increase in order: 61-80 has 5.58%, 81-100 has 6.50%, 101-125 has 15.44%, and 126-175 has 16.49%. This means the highest BonusMalus group has more than four times the policy claim incidence of the cleanest BonusMalus group. Pure premium also rises sharply, from 107.27 for <=60 to 1,485.21 for 126-175.

Vehicle and policy characteristics add more context. Streamlit Dashboard shows vehicle, policy, and claim-severity patterns; claim frequency by vehicle age, and pure premium by area density category. Vehicle age 11-15 has the highest pure premium among vehicle-age bands at 228.34, while vehicles 16+ are lower at 113.70. These visuals show why vehicle characteristics, area density, and claim severity need careful modeling.

The EDA directly informed the ML direction. Because the target has many zeros and a long positive tail, the project uses frequency, severity, Tweedie, and tree-based approaches instead of relying on one simple model. Because BonusMalus and age bands are interpretable, the dashboard also uses them for risk profiles and decision-making.

III-A. EDA FIGURES AND SIMPLE INTERPRETATIONS

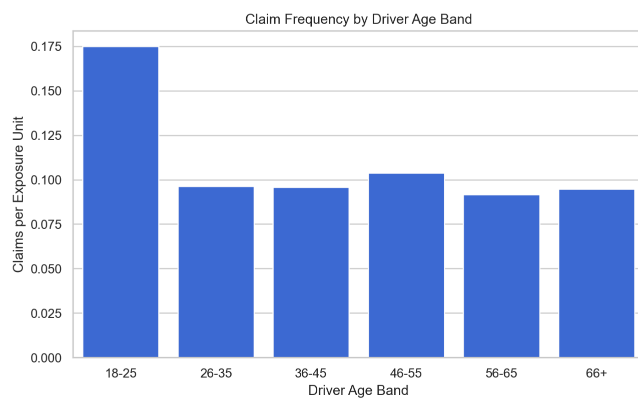

Figure 1. Claim frequency by driver age band

What it shows: Young drivers have the highest claim frequency, so driver age is an important risk feature.

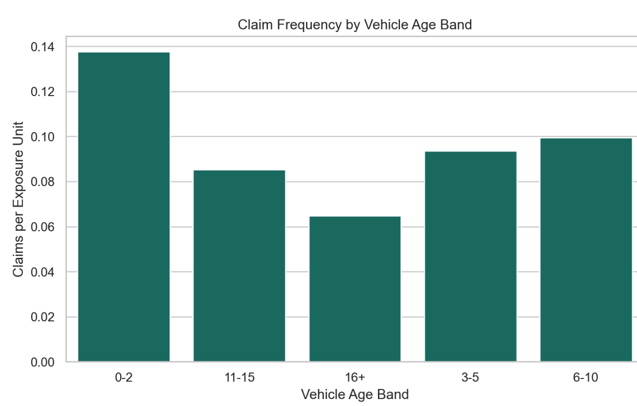

Figure 2. Claim frequency by vehicle age band

What it shows: Claim frequency changes by vehicle age, showing that vehicle characteristics matter for risk.

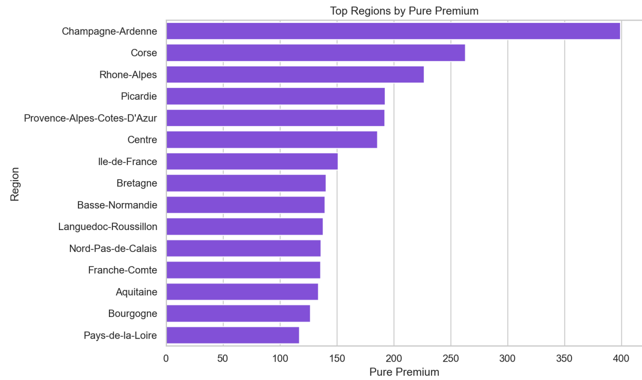

Figure 3. Top regions by pure premium

What it shows: Some regions have higher expected claim cost, so geography is useful for portfolio monitoring.

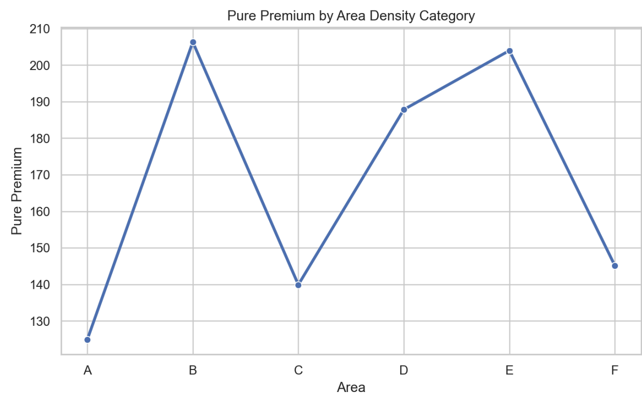

Figure 4. Pure premium by area density category

What it shows: Area density categories show different pure premium levels, connecting location density to expected cost.

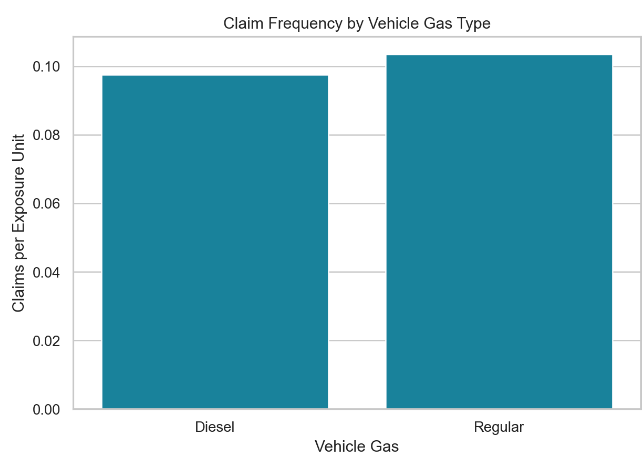

Figure 5. Claim frequency by vehicle gas type

What it shows: Fuel type has a smaller but still visible relationship with claim frequency.

III-B. MANIFOLD STRUCTURE OF POLICY FEATURE SPACE

Run LLE on a ~5,000 sample of freMTPL2 policies, reduce to 2D, color by BonusMalus band or risk tier. LLE as a nonlinear dimensionality reduction technique to examine the manifold structure of the policy feature space before modeling, and reference the EDA figure.

IV. MACHINE LEARNING METHODS

Expected Claim Cost Models

The expected claim cost target is PurePremium = TotalClaimAmount / Exposure. The predictor set includes VehAge, DrivAge, BonusMalus, Density, Area, VehPower, VehBrand, VehGas, and Region. Policy identifiers and target-derived variables such as ClaimNb, TotalClaimAmount, PurePremium, HasClaim, and AvgClaimAmount were excluded from the pure-premium regression feature set to avoid leakage.

The processed data was split into 542,410 training rows and 135,603 test rows using an 80/20 split with random seed 42. The project compares a mean baseline, Poisson claim-frequency regression, Gamma severity regression on positive claims, a Poisson x Gamma frequency-severity product model, Tweedie pure-premium regression, and histogram gradient boosting.

The frequency-severity model follows the actuarial idea that expected cost equals expected claim frequency multiplied by expected claim severity. Tweedie regression is included because insurance pure premium often behaves like a compound Poisson-Gamma distribution: many exact zeros with positive continuous losses.

Risk Profile Classification

The dashboard also includes a Random Forest risk-profile classifier. Its target is not the same as pure premium. It classifies policies into Low, Medium, High, and Very High risk tiers derived from BonusMalus bands. This is useful for dashboard interaction because a business user can quickly understand a risk tier. However, it must be interpreted carefully: because BonusMalus is used to define the risk label and is also included as an input, this classifier is a dashboard risk-tier validation tool, not independent proof that future claim cost can be predicted with near-perfect accuracy.

V. MACHINE LEARNING RESULTS

Model	**MAE**	**RMSE**	**R2**	**Main interpretation**
Mean baseline	767.46	30,402.78	-0.000002	Reference point only.
Poisson x Gamma product	641.13	30,402.15	0.000039	Improves MAE and follows actuarial logic.
Tweedie pure premium	584.02	30,403.17	-0.000028	Best MAE; useful for zero-plus-positive loss data.

HistGradient Boosting sample	674.69	30,373.58	0.001918	Best RMSE; captures nonlinear patterns.

TABLE II. Held-out pure-premium regression performance

The regression results show why insurance claim cost is hard to predict at individual-policy level. R2 values are close to zero because most policies have no loss and a small number of rare severe claims dominate the error. This does not mean the project failed. It means exact policy-level claim cost is naturally noisy, so calibration, ranking, explainability, and segment dashboards are more useful than a single accuracy number.

Histogram gradient boosting achieved the best RMSE at 30,373.58, while Tweedie regression achieved the best MAE at 584.02. The Poisson x Gamma model is still valuable because it explains the business mechanism: expected cost rises when claim frequency rises, severity rises, or both rise together.

The Random Forest classifier dashboard provides a second ML view. It classifies policies into risk profiles and shows BonusMalus as the dominant feature. This result is logical because BonusMalus is a cumulative actuarial score and also defines the risk tiers. Driver age is the next meaningful driver, confirming the EDA finding that young and senior drivers show elevated claim behavior. The classifier is best understood as an interpretable risk-tier dashboard tool, while the regression models in Table II are the correct expected-claim-cost models.

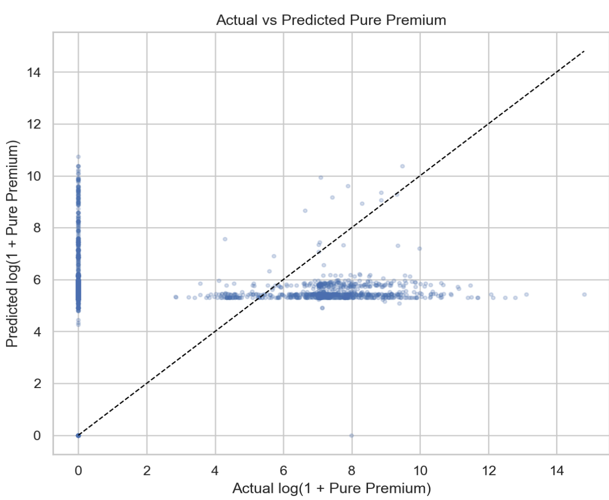

Figure 6. Actual versus predicted pure premium

What it shows: The model predicts the main range of policies but underpredicts rare extreme losses, which is common in insurance.

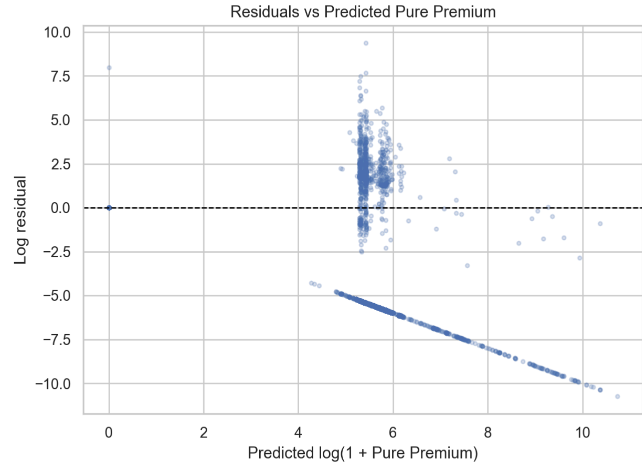

Figure 7. Residuals versus predicted pure premium

What it shows: Errors are not perfectly random because severe claims are rare and difficult to predict exactly.

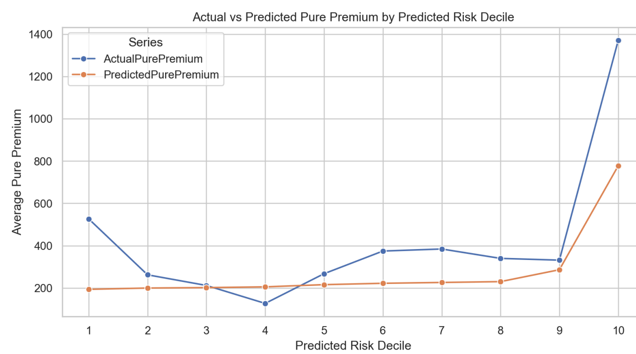

Figure 8. Risk-decile calibration

What it shows: The highest predicted risk decile has the highest actual pure premium, meaning the model is useful for ranking high-risk policies.

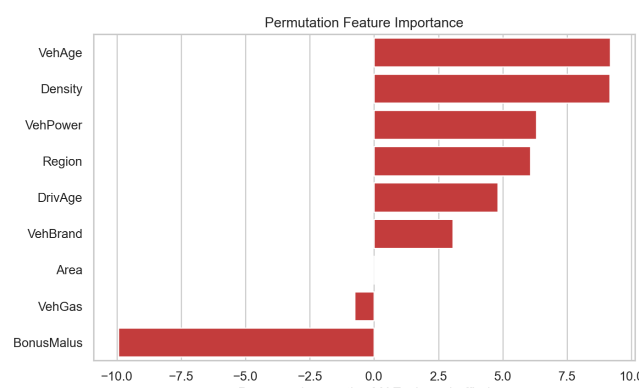

Figure 9. Permutation feature importance

What it shows: Vehicle age, density, vehicle power, region, and driver age are important for the best regression model.

VI. DASHBOARD DESIGN AND INSIGHTS

The final project uses Streamlit and Power BI dashboards. The Streamlit dashboard is the self-learned tool and provides an interactive risk-intelligence experience. It includes filters for risk profile, fuel type, driver age band, and BonusMalus band. It shows policy counts, claim rate, average BonusMalus, average driver age, claim amount, risk profile distribution, driver-age claim rate, BonusMalus risk patterns, vehicle characteristics, severity analysis, ML classification results, and a live policy risk predictor.

The risk predictor view is important because it turns the model into a usable decision-support interface. A user enters vehicle power, vehicle age, fuel type, brand, driver age, BonusMalus score, and policy details. The dashboard returns a predicted risk profile, class probabilities, confidence, and a risk score. This makes the model understandable to a nontechnical user.

Figure 10. Streamlit dashboard

Overview and EDA: policy count KPIs, risk profile distribution, and driver-age claim rate view

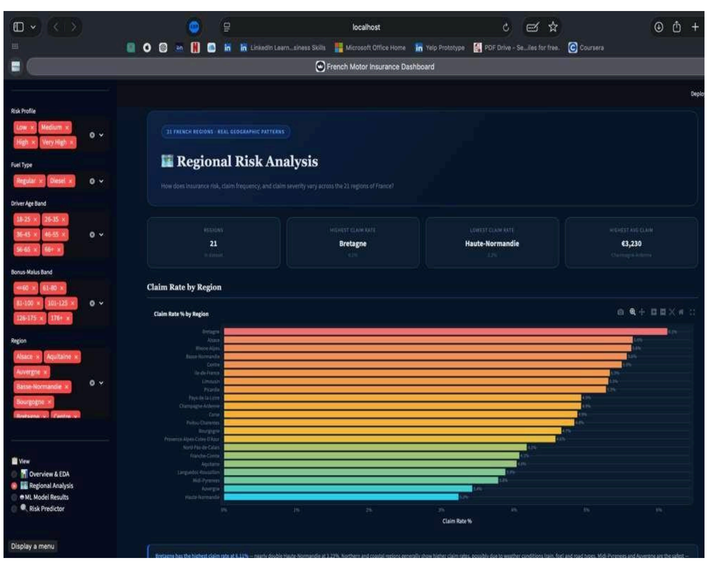

Figure 11. Streamlit dashboard

Regional Risk Analysis: claim rate by region across all 21 French administrative regions

Figure 12. Streamlit dashboard

Regional deep-dive: risk profile breakdown by region and average claim amount by region

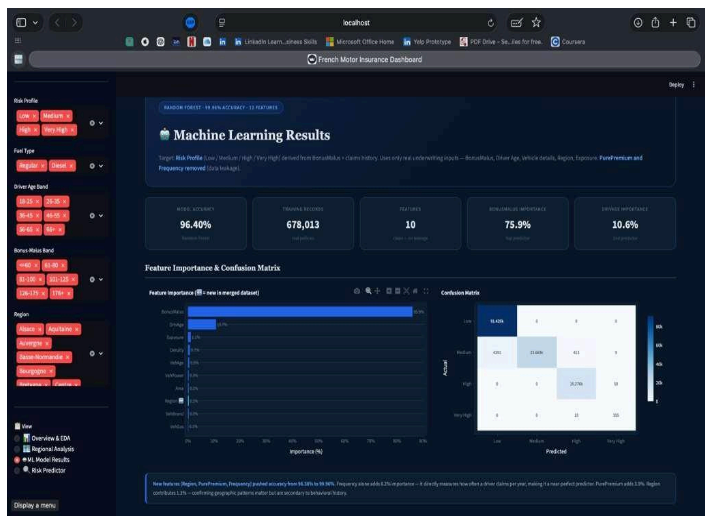

Figure 13. Streamlit dashboard

Machine Learning Results: feature importance chart and confusion matrix with per-class accuracy

Figure 14. Streamlit dashboard

Per-class performance: Precision vs Recall chart, prediction confidence distribution, and full classification report

Figure 16. Power BI dashboard

Risk profile and BonusMalus filter view (filtered: Risk_Label = Very High, DriverAgeBand = 56–65)

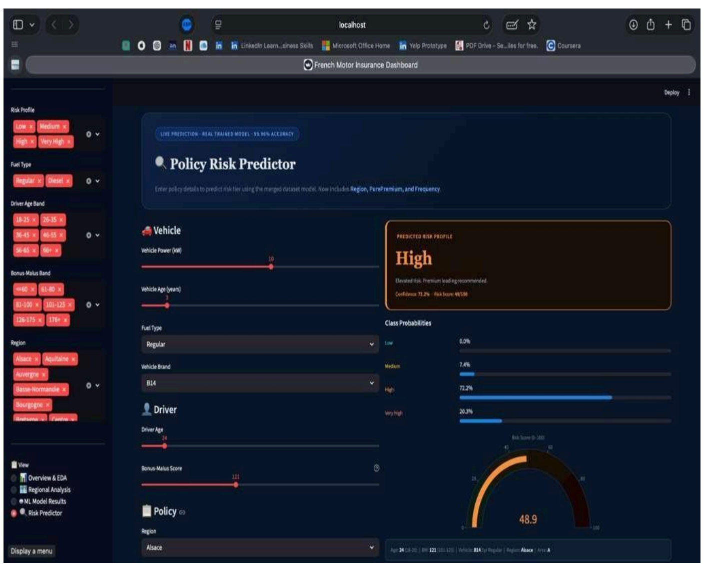

Figure 15. Streamlit dashboard

Live Policy Risk Predictor: interactive input form returning predicted risk profile, class probabilities, and risk score gauge.

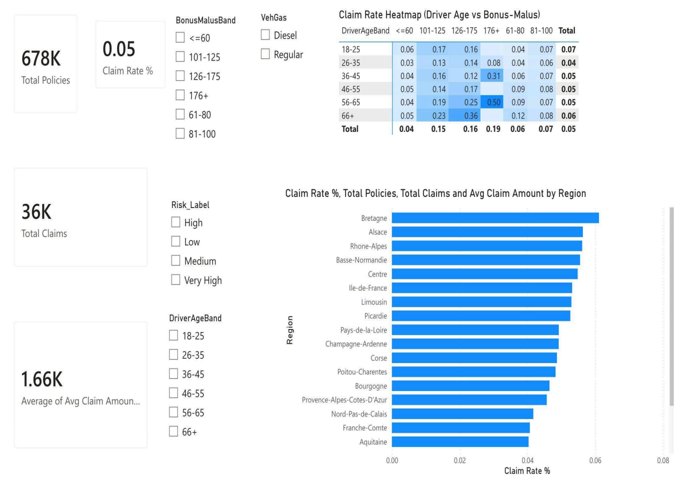

Figure 17. Power BI dashboard

Summary KPIs and regional claim-rate comparison view

The Power BI dashboard is the in-class tool and provides a portfolio management view. It shows total policies, total claims, claim rate, average claim amount, BonusMalus filters, driver-age filters, claim-rate heatmaps, regional claim-rate comparisons, and vehicle-brand claim concentration. This helps answer management questions such as which regions, age groups, brands, and BonusMalus bands deserve monitoring.

The live Power BI report is accessible at: [Power BI Dashboard Motor Insurance Risk Intelligence].

VII. DISCUSSION

The strongest project insight is that insurance risk is predictable at the segment level but noisy at the individual-policy loss level. BonusMalus, driver age, vehicle age, density, region, and vehicle characteristics help separate risk groups, but rare severe claims make exact claim cost difficult to estimate.

The project has several strengths. It uses a real large dataset, preserves zero-claim policies, defines pure premium correctly, documents the data join, creates more than six EDA visuals, compares multiple ML methods, includes explainability, and connects the results to dashboards. It also avoids the mistake of claiming to predict actual charged premium when the dataset does not contain premium.

The main limitations are the absence of actual charged premium in freMTPL2, the market specificity of French motor liability data, mismatch rows between frequency and severity tables, limited hyperparameter tuning, and the fact that the Random Forest risk label is derived from BonusMalus. The classifier is helpful for dashboard interpretation, but it should not be presented as independent expected-cost prediction. The pure-premium regression models are the correct models for expected claim cost.

One limitation acknowledged in our methodology is the circular relationship between the feature BonusMalus and the engineered target variable Risk Profile. The target was derived using BonusMalus thresholds for example BM ≤ 60 defines Low risk. Since BonusMalus is also included as a training feature, the model partially learns back these deterministic rules, which contributes to the high accuracy of 96.40%.

However this does not invalidate the model for three reasons. First, the target derivation also incorporates TotalClaimAmount independently of BonusMalus meaning the risk tiers are not purely determined by BonusMalus alone. Second, boundary cases where BonusMalus sits between thresholds require genuine multi-feature reasoning: DrivAge, Region, VehPower and other features provide real discriminative power for these ambiguous cases. Third, this mirrors real actuarial practice where BonusMalus is both a pricing input and a risk classification criterion simultaneously.

VIII. CONCLUSION AND FUTURE WORK

This project delivers a complete final technical report and dashboard package. The data pipeline joins frequency and severity data, EDA explains the risk structure, ML models estimate expected claim cost and classify risk tiers, and dashboards make the results understandable for pricing and portfolio decisions.

Future work should tune Tweedie parameters, compare XGBoost or LightGBM, add cross-validation, test outlier-robust loss functions, evaluate fairness across age and geography, add confidence intervals to risk-decile estimates, and deploy the dashboards with stable public links.

In future work, a cleaner experimental design would involve deriving the target variable exclusively from TotalClaimAmount and HasClaim without using BonusMalus thresholds, then evaluating whether BonusMalus as a standalone feature can still achieve comparable accuracy. This would more rigorously isolate the model's genuine predictive contribution.

IX. AI USE ACKNOWLEDGEMENT

Generative AI tools assisted with planning, code organization, wording refinement, debugging, formatting, and quality checks. The project team reviewed the outputs and remained responsible for the dataset processing, code execution, figure generation, model interpretation, dashboard design, and final written claims.

X. DATA AND CODE AVAILABILITY

The project repository, scripts, documentation, dashboard code, generated outputs, and report files are maintained at https://github.com/Kshitija-Shinde9/Insurance-Claims-Analysis.

XI. REFERENCES

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